Dorey's Resource Book

Topic 3 – Right-angled triangles

Contents

Episode 1. Pythagoras' theorem

1 Pythagoras' theorem

ANCHOR QUESTION

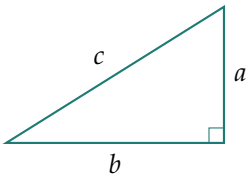
A right-angled triangle has shorter sides 9 and 12. Find the length of the hypotenuse.

SYLLABUS MST-12-S1-04

- Apply Pythagoras' theorem to solve practical problems in two dimensions

NOTES

- The **hypotenuse** is the longest side of a right-angled triangle. It is opposite the right angle.
- **Pythagoras' theorem:** $c^2 = a^2 + b^2$, where c is the hypotenuse.
- To find the hypotenuse, add the squares: $c = \sqrt{a^2 + b^2}$.
- To find a shorter side, subtract the squares: $a = \sqrt{c^2 - b^2}$.
- Write all sides in the same units first. Round only at the end.



WORKED EXAMPLE

A right-angled triangle has shorter sides 8 and 15. Find the length of the hypotenuse, x.

$x^2 = 8^2 + 15^2$ hypotenuse: add the squares

$x^2 = 64 + 225 = 289$

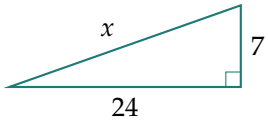
$x = \sqrt{289} = 17$ square root

Tier 1 · Finding the hypotenuse

1. The shorter sides of a right-angled triangle are 3 and 4. Find the hypotenuse.

2. The shorter sides of a right-angled triangle are 5 and 12. Find the hypotenuse.

3. Find x.



Tier 2 · Rounding

4. The shorter sides of a right-angled triangle are 4 and 9. Find the hypotenuse, correct to two decimal places.

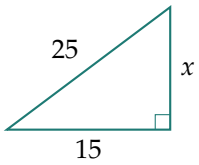
5. The shorter sides of a right-angled triangle are 2.5 and 6. Find the hypotenuse.

6. The shorter sides of a right-angled triangle are 11 and 14. Find the hypotenuse, correct to one decimal place.

Tier 3 · Finding a shorter side

7. A right-angled triangle has a hypotenuse of 10 and one shorter side of 8. Find the other shorter side.

8. Find x.



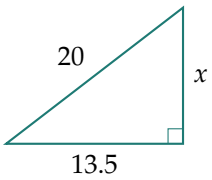
9. A right-angled triangle has a hypotenuse of 17 and one shorter side of 8. Find the other shorter side.

Tier 4 · Shorter side with rounding

10. A right-angled triangle has a hypotenuse of 12 and one shorter side of 7. Find the other shorter side, correct to two decimal places.

11. A right-angled triangle has a hypotenuse of 8.5 and one shorter side of 4.2. Find the other shorter side, correct to two decimal places.

12. Find x, correct to two decimal places.



Tier 5 · Mixed units

13. The shorter sides of a right-angled triangle are 60 cm and 0.8 m. Find the hypotenuse in metres.

14. A right-angled triangle has a hypotenuse of 2.6 m and a shorter side of 100 cm. Find the other side in metres.

15. The shorter sides of a right-angled triangle are 1.5 km and 800 m. Find the hypotenuse in kilometres.

Tier 6 · Two triangles

16. In triangle ABC, D is a point on BC and AD is perpendicular to BC. $AB = 13$, $BD = 5$ and $DC = 16$. Find AD, then find AC.

17. In triangle ABC, AD is perpendicular to BC. $AB = 10$, $BD = 6$ and $DC = 6$. Find AC.

18. A square has sides of 7 cm. Find the length of its diagonal, correct to two decimal places.

Conceptual

19. Use Pythagoras' theorem to explain why the hypotenuse is always the longest side.

20. A triangle has a hypotenuse of 10 and a shorter side of 6. Mia finds the other side as $\sqrt{100 + 36} = 11.66$. Explain her mistake and find the correct answer.

21. A square has a diagonal of 10 cm. Find the side length, correct to two decimal places.

22. Explain why the sides must be in the same units before using Pythagoras' theorem.

23. A composite shape is made of a right-angled triangle with shorter sides 9 cm and 12 cm, and a semicircle drawn on the hypotenuse.

(a) Find the diameter of the semicircle. (2 marks)

(b) Find the perimeter of the composite shape, to the nearest centimetre. (2 marks)

Revision

24. A spinner with 5 equal sections numbered 1 to 5 is spun and a coin is tossed.

(a) How many outcomes are there? (1 mark)

(b) Find $P(\text{head and an even number})$. (1 mark)

(c) Find the probability of NOT getting a head and an even number. (1 mark)

(d) Find the expected number of these results in 300 goes. (2 marks)

25. A garden bed is a trapezium with parallel sides of 6 m and 9 m, which are 4 m apart.

(a) Find the area of the garden bed. (2 marks)

(b) Mulch is spread 0.1 m deep. Find the volume of mulch needed. (1 mark)

(c) Mulch costs \$65 per m^3, plus a \$45 delivery fee. Find the total cost. (2 marks)

26. A tap fills a 9000 L tank at a rate of 15 L/min.

(a) How many hours does it take to fill the tank? (2 marks)

(b) Convert 15 L/min to L/h. (1 mark)

(c) Water costs \$3.20 per kilolitre. Find the cost of the water to fill the tank. (2 marks)

Solutions

1. $\sqrt{9+16} = \sqrt{25} = 5$

2. $\sqrt{25+144} = \sqrt{169} = 13$

3. $\sqrt{49+576} = \sqrt{625} = 25$

4. $\sqrt{16+81} = \sqrt{97} = 9.848\ldots \approx 9.85$

5. $\sqrt{6.25+36} = \sqrt{42.25} = 6.5$

6. $\sqrt{121+196} = \sqrt{317} = 17.80\ldots \approx 17.8$

7. $\sqrt{100-64} = \sqrt{36} = 6$

8. $\sqrt{625-225} = \sqrt{400} = 20$

9. $\sqrt{289-64} = \sqrt{225} = 15$

10. $\sqrt{144-49} = \sqrt{95} = 9.746\ldots \approx 9.75$

11. $\sqrt{72.25-17.64} = \sqrt{54.61} = 7.389\ldots \approx 7.39$

12. $\sqrt{400-182.25} = \sqrt{217.75} = 14.756\ldots \approx 14.76$

13. 0.6 m and 0.8 m. $\sqrt{0.36+0.64} = 1$ m

14. $\sqrt{6.76-1} = \sqrt{5.76} = 2.4$ m

15. 1.5 km and 0.8 km. $\sqrt{2.25+0.64} = \sqrt{2.89} = 1.7$ km

16. $AD = \sqrt{169-25} = 12$. $AC = \sqrt{144+256} = \sqrt{400} = 20$

17. $AD = \sqrt{100-36} = 8$. $AC = \sqrt{64+36} = 10$

18. $\sqrt{49+49} = \sqrt{98} = 9.899\ldots \approx 9.90$ cm

19. $c^2 = a^2 + b^2$, so c^2 is bigger than a^2 and bigger than b^2. That means c is longer than both a and b.

20. She added instead of subtracting. A shorter side cannot be longer than the hypotenuse. $\sqrt{100-36} = 8$.

21. $s^2 + s^2 = 100$, so $s^2 = 50$ and $s = \sqrt{50} = 7.071\ldots \approx 7.07$ cm.

22. The theorem adds and subtracts the squares of the lengths, which only makes sense when they are measured in the same unit.

23. (a) $\sqrt{81+144} = \sqrt{225} = 15$ cm. (b) Arc $= \frac{1}{2} \times \pi \times 15 = 23.56\ldots$ cm. Perimeter $= 9 + 12 + 23.56 = 44.56 \approx 45$ cm.

24. (a) $5 \times 2 = 10$. (b) H2, H4: $\frac{2}{10} = \frac{1}{5}$. (c) $1 - \frac{1}{5} = \frac{4}{5}$. (d) $300 \times \frac{1}{5} = 60$.

25. (a) $\frac{4}{2} \times (6+9) = 30$ m^2. (b) $30 \times 0.1 = 3$ m^3. (c) $3 \times \$65 + \$45 = \$240$.

26. (a) $9000 \div 15 = 600$ min $= 10$ h. (b) $15 \times 60 = 900$ L/h. (c) 9000 L $= 9$ kL. $9 \times \$3.20 = \$28.80$.